

The Architectural Solution (The 2x2 Framework)

■■ Thursday, Sept 24, 2026: The Architectural Solution (The 2x2 Framework)

Next time a team proposes an AI feature, run it through this simple 2x2 decision framework.

We need clear, objective boundaries to keep our production architectures lean. When evaluating a new feature, do not ask "Can we use AI?" Ask: **What is the structure of our input data, and what is the nature of the expected output?**

If you are new to data engineering, think of the inputs in two simple buckets:

- **Structured Data:** Tidy, clean data that fits perfectly into database rows (like user ages, timestamps, transaction amounts, or product IDs).
- **Unstructured Data:** Raw, unformatted real-world files that do not have a predefined format (like free-text emails, call transcripts, images, or dense PDF files).

Here is how you execute this matrix across your microservices at a high level:

Quadrant 1 (Structured Input + Predictive Output) → Financial Fraud Engines:

Strictly Traditional ML. Deploy local models like XGBoost. Operating in single-digit milliseconds at near-zero incremental cost. Do not use an LLM.

Quadrant 2 (Unstructured Input + Categorical Output) → Ticket Routing:

Start with Traditional ML. A small, locally hosted text classification model (like a distilled BERT instance) handles text formatting for fractions of a penny.

Quadrant 3 (Structured Input + Generative Output) → Automated Report Writing:

Hybrid Systems. Aggregate your numbers via a traditional data pipeline first, then feed the final numbers to an LLM strictly to polish the prose.

Quadrant 4 (Unstructured Input + Generative Output) → Legal Contract Synthesis:

Generative AI. This is where foundational LLMs are fully justified because the task requires deep linguistic context and dynamic text synthesis.

Here is how the data type dictates your underlying infrastructure choice:

Predictive vs. Generative AI Matrix by Data Type

[EXPECTED OUTPUT TYPE]

Predictive / Labels

| Generative / Synthesis

DATA TYPE

STRUCTURED DATA

QUADRANT 1

Traditional ML

Credit Scoring, Churn Prediction



QUADRANT 3

Hybrid Systems

Automated Financial Report Writing



QUADRANT 2

Traditional ML

Support Ticket Routing (BERT)



Routing



QUADRANT 4

Generative AI

Contract Summaries & Developer Copilots



Contract Summaries



Developer Copilot



[Blue = Traditional ML / Predictive] [= Generative AI / Synthesis]